

Giuseppe Vitolo

SOFTWARE ENGINEERING | BACKEND SYSTEMS | B2B PRODUCT DATA

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GitHub | LinkedIn | Portfolio | Pokoin

Computer Engineering MSc student at Aarhus University building backend services, data pipelines and web products with Rust, Kubernetes and PostgreSQL. Experience processing 52.9 million historical price observations and connecting supplier documentation, product information and retail workflows through a B2B internship.

TECHNICAL SKILLS

Backend & data: Rust (Axum, Tokio, SQLx), Python, Java, SQL, PostgreSQL, MySQL, REST APIs

Infrastructure & web: Kubernetes (k3s), Docker, Linux, Git, Cloudflare, React, JavaScript, HTML/CSS

ENGINEERING & PROFESSIONAL EXPERIENCE

Backend & Data Engineering

2026

Pokoin - independent software project

- Built marketplace APIs and React interfaces; shipped autocomplete in Rust (Axum, Tokio, SQLx) and deployed the website on Cloudflare.
- Reduced home-feed median latency from **6.9 s to 26 ms** and raised throughput from **1.3 to 284 requests/s** using edge caching and request coalescing (September 2026 test, 8 concurrent requests).
- Configured Kubernetes (k3s) API overflow, health-based failover and PostgreSQL writer/replica separation for consistent writes and distributed reads.
- Imported **52.9 million historical price observations** from 951 archives into 32 PostgreSQL monthly partitions, validating row counts and preserving source hashes.
- Automated daily snapshots of **624,798 product/variant rows** and **510,593 distinct products**; linked provider data to retail-facing catalog and pricing APIs through transactional, retryable imports.

B2B Software Development Internship

2026

product

- Mapped product-data handoffs, documentation ownership and data quality across suppliers, companies and retailers for Digital Product Passport readiness.
- Built an interactive assessment website with purchasing and sales paths, product-data mapping and actionable results; iterated the sprint prototype with stakeholder feedback.

Java Software Developer - Internship

Jan-Mar 2025

Innovation Engineering, Salerno, Italy

- Built JavaFX dashboards for **3 simulated industrial machines**, integrating Modbus TCP acquisition with timestamped measurements in MySQL.

Communications Assistant - Digital Education

Jul 2025-Feb 2026

Tmelnik, Prague, Czech Republic

- Coordinated digital communication, event activities and reporting with international youth-education partners.

EDUCATION

MSc Computer Engineering

2026-2028 (expected)

Aarhus University, Denmark · Distributed Systems and Security; Modelling and Verification; Software Correctness.

BSc Computer Engineering

2025

University of Salerno, Italy · Programming, algorithms, operating systems, networks, databases and software engineering.

LANGUAGES & CERTIFICATES

Languages: Italian (native), English (C1), Danish (studying toward B1, A2B Sprogskole).

Certificates: Cambridge C1 Advanced; Docker Foundations; Google Git and GitHub; IBM Cybersecurity Fundamentals; AWS Cloud Technical Essentials.